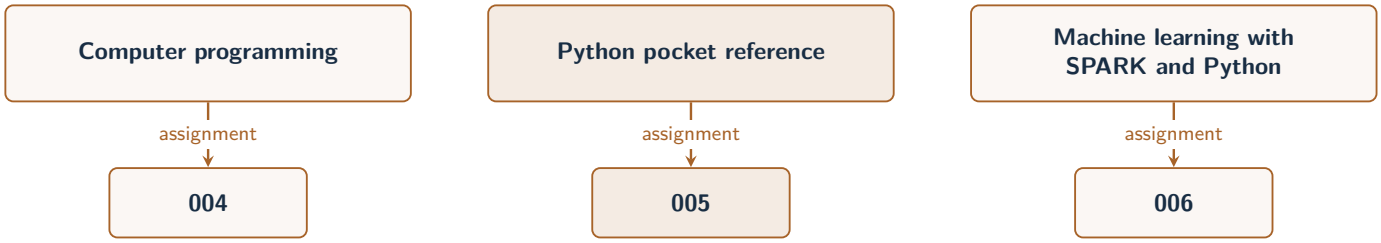
KOS Computing Structures

One computing theme, with local context. Tables precede their corresponding diagrams.

01 DDC: observed labels and document descriptions

Labels are corpus assignments. Descriptions below are document-text excerpts, not official class definitions.

Label	Document title	Description (excerpt)	D / A row
004	Computer programming	There are some unobvious dead lock problems caused by inappropriate acquiring and releasing resources among current transactions. At first, this...	5384
005	Python pocket reference	Updated for both Python 3.4 and 2.7, this guide provides concise information on Python types and statements, special method names, built-in...	6303
006	Machine learning with SPARK and Python	Machine Learning with Spark and Python Essential Techniques for Predictive Analytics, Second Edition simplifies ML for practical uses by...	6873

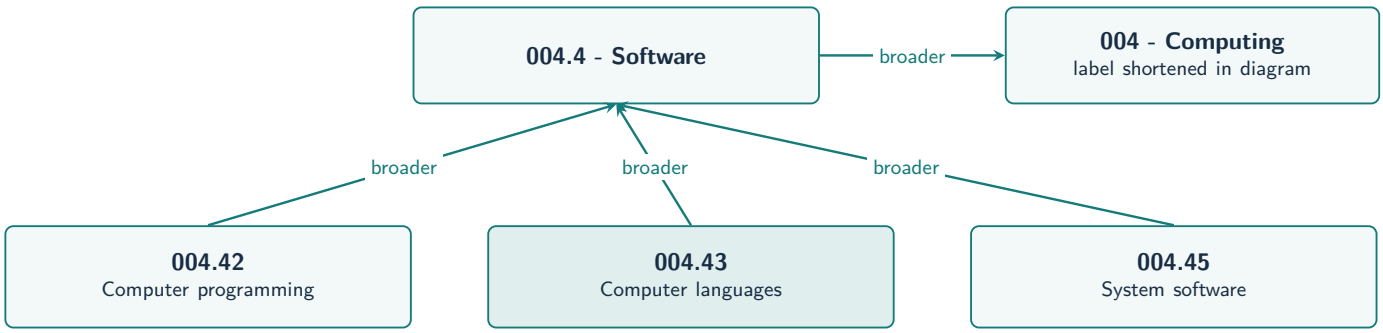


004, 005 and 006 are neighboring observed codes only. No common parent or official caption is supplied.

D / A: DDC_Prepared_Documents_001.xlsx / DDC_Prepared_Assignments_001.xlsx, Documents / Assignments; rows above. Labels: DDC_Prepared.xlsx, Units 6-8.

02 UDC: neighboring concepts with a shared broader concept

Notation	Preferred label	Broader	Source rows
004	Computer science and technology. Computing. Data processing	context only	U657
004.4	Software	004	U678 / S3358
004.42	Computer programming. Computer programs	004.4	U683 / S3384
004.43	Computer languages	004.4	U684 / S3388
004.45	System software	004.4	U685 / S3392



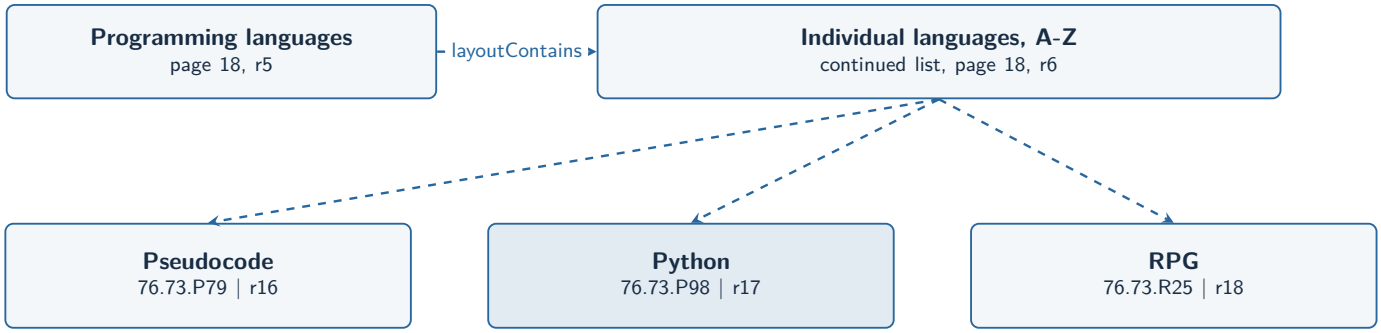
All arrows: skos:broader, from specific to broader. Source: UDC_Summary_Prepared.xlsx; U = Units, S = Statements. The table preserves the full preferred labels.

Schedule Context and Subject Headings

A shared level requires a recorded common parent; adjacency alone does not establish hierarchy.

03 LCC: Python and adjacent entries in the printed schedule

Row	Printed notation	Label	Units row
r5	(heading)	Programming languages	4652
r6	(heading)	Individual languages, A-Z – Continued	4653
r16	76.73.P79	Pseudocode	4663
r17	76.73.P98	Python	4664
r18	76.73.R25	RPG	4665

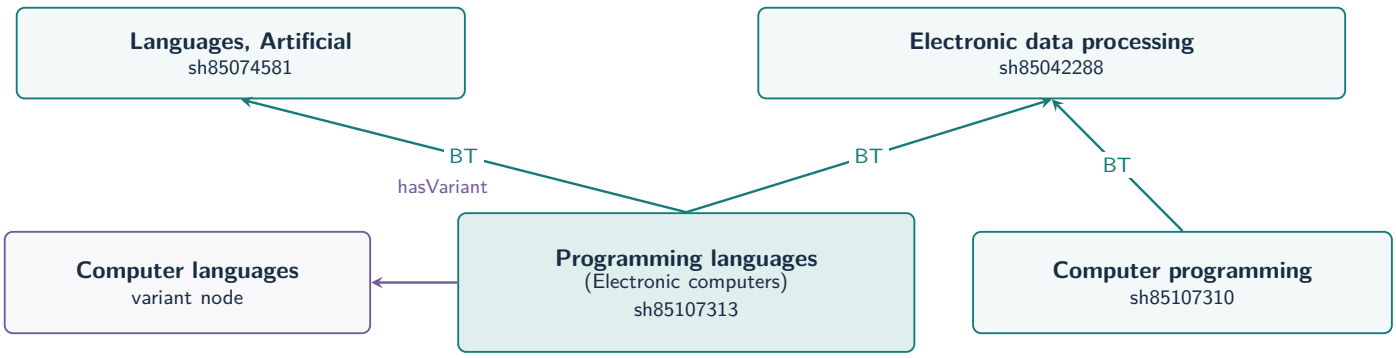


Dashed edges are layoutContains derived from indentation, not native is-a. Source: LCC_Prepared.xlsx, Statements 4264 and 4274-4276. The entries share page 18, r6.

Python contextual notation: QA76.73.P98. Pseudocode is retained in the full local workbook, outside the 10,000-row subset.

04 LCSH: a shared broader heading and an alternative name

Subject label	Relation	Object label	Row
Programming languages (Electronic computers)	hasBroaderAuthority	Languages, Artificial	11986
Programming languages (Electronic computers)	hasBroaderAuthority	Electronic data processing	11724
Computer programming	hasBroaderAuthority	Electronic data processing	11650
Programming languages (Electronic computers)	hasVariant	Computer languages	11787



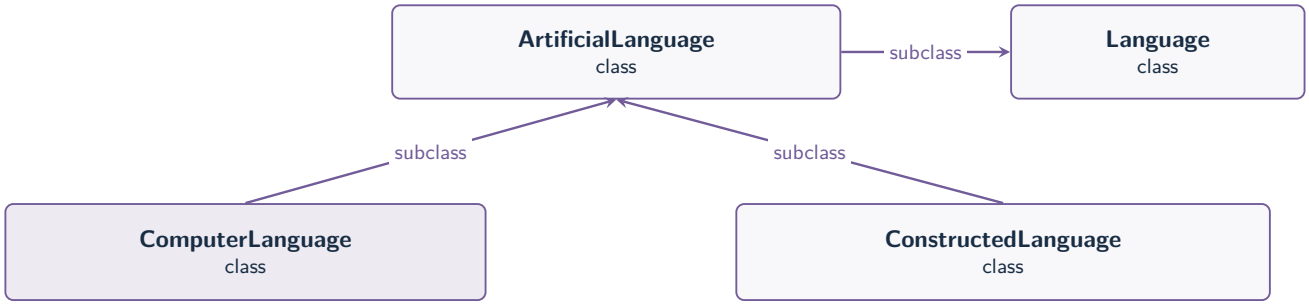
BT = hasBroaderAuthority. Source: LCSH_Statements_Part007.xlsx, Statements rows above. Variant text: LCSH_Texts_Part002.xlsx, Texts 12504. A variant is not a sibling class.

Formal Classes and Wikidata Statements

The arrows retain their source meaning. A common parent provides the local sibling context.

05 SUMO: classes, a shared superclass and a native rule

Class	Relation	Superclass	Logic row
ComputerLanguage	subclass	ArtificialLanguage	457
ConstructedLanguage	subclass	ArtificialLanguage	466
ArtificialLanguage	subclass	Language	455



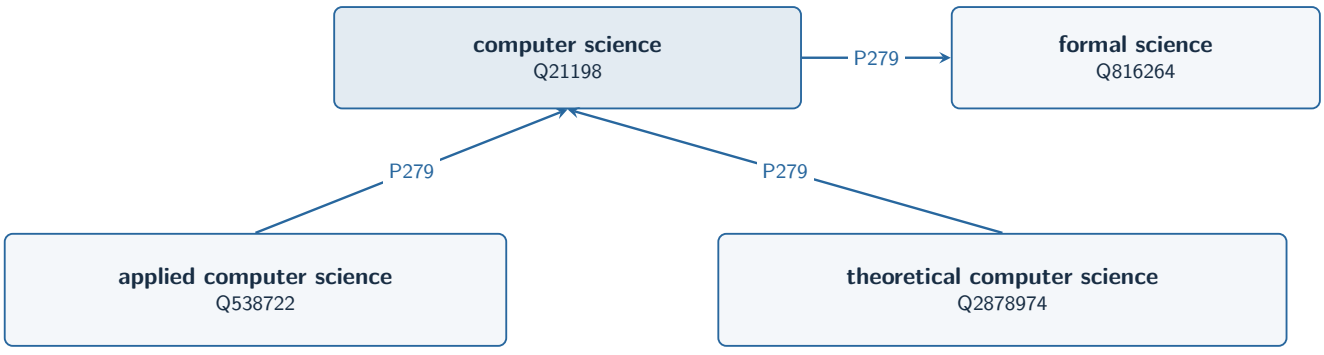
Native general rule (line breaks normalized):

(=> (and (subclass ?X ?Y) (instance ?Z ?X)) (instance ?Z ?Y))

An instance of a subclass is also an instance of its superclass. Source: SUMO_Prepared_Native_Logic.xlsx, Native_Logic rows above; general rule at row 26. No named instance is invented.

06 Wikidata: neighboring computer-science disciplines

Subject label	Relation	Object label	Row
applied computer science	P279 (subclass of)	computer science	72803
theoretical computer science	P279 (subclass of)	computer science	51952
computer science	P279 (subclass of)	formal science	8710



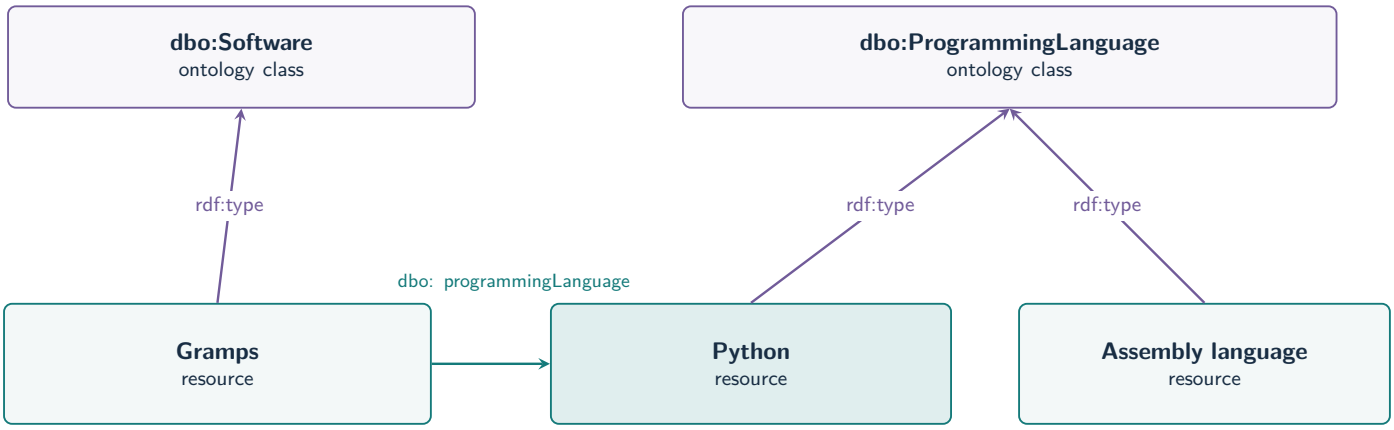
P279 = subclass of. Source: Wikidata_Prepared_Statements_001.xlsx, Statements rows above. This illustrates the broader discipline level, not an exact Python-level match.

Typed Resources and Additional Candidates

One topic can appear as a book, subject, class or resource. These are not exact equivalence mappings.

07 DBpedia: resource facts and resources of the same type

Subject label	Relation	Object label	Row
Gramps (software)	dbo:programmingLanguage	Python (programming language)	881
Gramps (software)	rdf:type	software	3103
Python (programming language)	rdf:type	programming language	3583
Assembly language	rdf:type	programming language	2744



Python and Assembly language share a type; neither is drawn as a subclass of the other. Source: DBpedia_Prepared.xlsx, Statements rows above.

dbo: = dbpedia.org/ontology/; resources use dbpedia.org/resource/Gramps_(software), Python_(programming_language) and Assembly_language.

08 Additional candidates: current project data status

Candidate	Status	Meaning in this project
YAGO 4.6	Excel ready	10,000 taxonomy relations; structure example pending.
FAST	Excel ready	10,000 semantic statements; structure example pending.
BabelNet	No data	No prepared Excel sample is included in this comparison.
WordNet	Deferred	Lower priority; no separate structure sample is included.

Excel ready means a separate cleaned workbook is available. It does not mean a structure diagram has been added. No data refers only to this project.

Scope: the seven illustrated KOS use the retained local collection. Some neighbors are outside the 10,000-row subsets. FAST and YAGO 4.6 workbooks are separate additions; their diagrams are pending.